

Bugs found: 6

1. Pointer arithmetic results in a pointer 'q' pointing outside the bounds of the allocated memory block 'p'.
2. Writing to '*q' causes a buffer overflow as it accesses memory outside the allocated block.
3. Accessing '*p' after calling 'free(p)' is a use-after-free vulnerability.
4. The loop ' $i < 6$ ' exceeds the bounds of the array 'str', which has a size of 5.
5. The 'printf' call results in a buffer over-read because 'str' is not null-terminated.
6. The code fails to check if 'malloc' returned NULL before dereferencing 'p'.